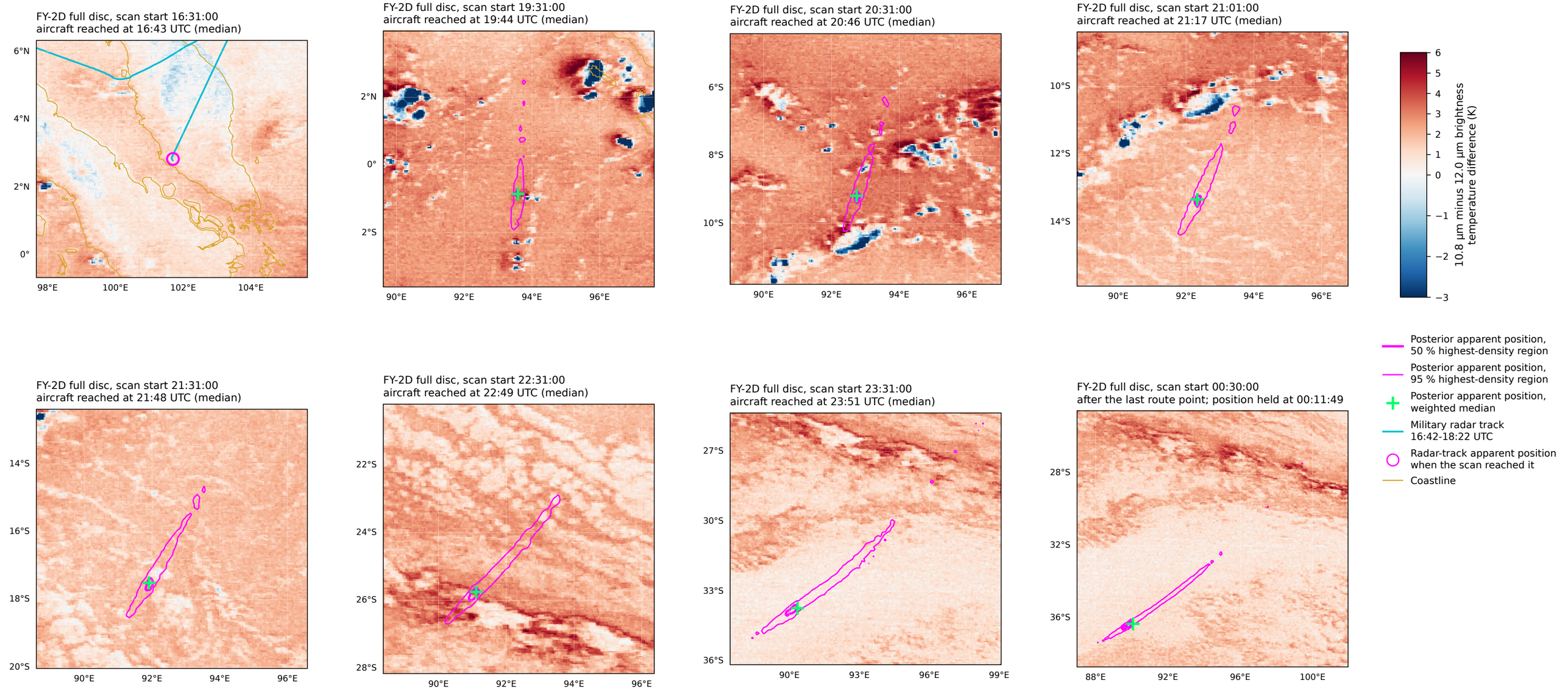
FY-2D split-window difference around the aircraft position at each archived scan, 7-8 March 2014



Plain note: Each panel shows, for one archived image, the difference between two infrared channels. Thin ice cloud and young condensation trails usually show as a stronger red than their surroundings. Clear sea is also red in the moist tropics, so a condensation trail would have to stand out as a thin straight line. The magenta lines show the most probable area for the aircraft. This chart is for inspection; it does not claim a detection.

Technical note: FY-2D (86.45 deg E, element set): full-resolution S-VISSR HRIT IR1 10.8 um / IR2 12.0 um as re-encoded by Chiba University CEReS (ftp://fy.cr.chiba-u.ac.jp/FY2-D/HRIT/2014/), GEOS(86.0) navigation from the file header, calibration from header type 3; nadir IR sampling 5 km; file time stamps are CEReS labels (line times in these files are placeholders, so scan times are modelled N->S 25 min from the label). Aircraft at FL350; 'apparent position' = where the line of sight through the aircraft meets the surface (parallax correction applied so the contours line up with a geolocated image). Posterior: core run J BTO+BFO, 148,000 routes, family weights waypoints 0.9982, free 0.0013, repro-radar 0.0003, descent-climb 0.0002; position at the modelled time the scan line reached each route; routes end 00:11:49 UTC, held there for later scans. Highest-density regions from a 0.1 deg weighted histogram. Before 18:01:49 the radar track (Geoscience Australia) is used. Split-window difference: thin ice cloud and young contrails are usually positive (red). Coastlines: Natural Earth 10 m. Altitude FL300-FL400 would shift the apparent positions by a further 0-15 % of the parallax shown in the sensor table.